

SECTION 07 SPECIFICATION FOR THERMAL & ACOUSTIC INSULATION

1. SCOPE

This specification covers the technical requirements and essential particulars for the supply, application and finishing of the composite thermal insulation for cold equipment, piping systems, air-conditioning ducts, etc. The scope of supply of the contractor shall include, but not be limited to, the following items:

- (a) Insulation material as specified
- (b) Finishing and cladding material as specified
- (c) Auxiliary materials such as binding and lacing wires, wire netting, bands, screws, pop-rivets, etc. as required
- (d) Angles, clamps, lugs, etc. for supporting insulation
- (e) Weather hoods
- (f) Any material as may be required for making the insulation complete

2. CODES AND STANDARDS

The following are some of the codes and standards relevant to this specification:

IS 277	Galvanised Steel Sheet (Plain and Corrugated)
IS 737	Wrought Aluminium and Aluminium alloy Sheet and Strip for General Engineering Purposes
IS 14164	Industrial Application and finishing of Thermal Insulation Materials at Temperatures above (-) 80° C and up to (+) 700° C
BS 5970	Thermal Insulation of Pipework and Equipment (in the Temperature Range (-) 100° C to (+) 870° C

3. MATERIAL

3.1. For outdoor installation

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber with UV treated woven glass cloth factory laminated on one side having thickness 7 mils/0.18 mm.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 0°C as per EN 12667.
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 7,000 without any external barrier as per EN 12086.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990. The Class O and 1 Properties should be for bare insulation material, i.e. without any type of cladding or lamination
- The insulation material shall have fire performance of V0, HB as per UL 94, 1996.
- The complete insulation system (NBR sheets with factory pre laminated GC) shall be FM (Factory Mutual), USA Approved.

Comply



NA

- Unlike other generics it should not drip or propagate fire and should be self-extinguishing
- Insulation should be approved by Gujarat state fire department
- Density of insulation material shall be between 40 to 55 Kg/m³.
- The insulation material shall be dust and fibre free.
- The insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of 0 Deg.C as per EN 14706.
- Glass Cloth Treatment: Shall be treated water based acrylic binder to give crisp and non-piling property to the fabric, to help in easy installation, minimize fiber erosion, good aesthetics and resistance to abrasion. Fibre spillage / Thread raveling shall be minimum.
- Glass Cloth : Weight shall be 205 +/- 10 grams per square meter, Tensile Strength shall be
- 275 +/- 25 Kg / 50 mm (minimum) and Thickness shall be 0.18 mm / 7 mil.
- Glass Cloth shall be UV resistant for artificial weathering as per EN ISO 4892-2 Method - A.
- GC Color - Green, Blue, Black
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Thickness of the insulation shall be as per manufacturer recommendation.
- For GC Overlap joints, silicon-based sealants to be used in case of exposure to UV Rays.
- Insulation manufacturer to ensure 1 full day free of cost application training to installer team. After that, inspection to be done by insulation manufacturer team on fortnightly basis to check the insulation workmanship and submit the reports. Inspection cost to be borne by contractor/installer.
- Tapes for the longitudinal joints should be as per the manufacturer's recommendation

Comply

Accessories - Adhesives

- Special adhesive for installing all flexible insulation materials Adhesive shall be used for bonding the seams securely together. All butt & longitudinal joints, fittings and coverings must be closed with adhesive.
- The adhesive shall be specially formulated for the Duct insulation application and supplied by insulation manufacturer. The adhesive shall be free from benzene.
- The adhesive shall have excellent bonding to porous and non-porous surfaces.

Technical Details

- Solvent Based Rubber contact adhesive
- Colour: Pale Yellow
- Temperature Range: +5°C to +85°C
- Form: Liquid
- Material: Rubber based contact adhesive in blend of solvents
- Minimum Drying Time (airing time): 4-5 minutes
- Setting Time: 36 hours
- Viscosity @ 30°C: 700 - 1200 cps
- Specific Gravity: 0.80 - 0.85 g/cc
- Density: 0.083 kg/m³ + 0.01
- Flash Point: > 10 °C
- Coverage: 6-8 square meters of bonded area per Litre (Bonded area - Area of insulation material and duct together)
- Tack Retention Time: 25 minutes

Comply



- Recommended Adhesives - Armaflex Ductseal / Pidilite SR 505

Tapes

Comply

- OEM quality 50 mm wide and 3 mm thick foam based insulation tapes of closed cell nitrile rubber with GC cloth to be used as per insulation material manufacturer's recommendations

3.2. For Indoor application_

Comply

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber with aluminum foil laminated on one side.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 0°C as per EN 12667.
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 60,000 as per EN 12086.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- Unlike other generics it should not drip or propagate fire and should be self-extinguishing
- Insulation should be approved by Gujarat state fire department
- Density of laminated insulation material shall be between 50 to 70 Kg/m³.
- Density of base foam insulation material shall be between 40 to 55 Kg/m³.
- The insulation material shall be dust and fibre free.
- The insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of 0 Deg.C as per EN 14706.
- The Aluminium foil (Plain / Embossed) shall be of 60 microns thickness and weight shall be 193 gsm as per EN ISO 2286-2.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Thickness of the insulation shall be as specified for the individual application.
- The insulation material shall be installed as per manufacturer's recommendation.
- Insulation manufacturer to ensure 1 full day free of cost application training to installer team. After that, inspection to be done by insulation manufacturer team on fortnightly basis to check the insulation workmanship and submit the reports. Inspection cost to be bearded by contractor/installer.



4. ACOUSTIC INSULATION

MATERIAL:

- Material shall be engineered Nitrile Rubber open cell foam
- The Random Incidence Sound Absorption Coefficient (RISAC); tested as per ISO 354, shall be minimum as per enclosed chart
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Freq (Hz)	125	250	500	1000	2000	4000	NRC
10 mm	0.03	0.04	0.14	0.04	0.88	1.00	0.35
15 mm	0.01	0.09	0.29	0.74	1.08	0.83	0.55
20 mm	0.04	0.13	0.4	0.9	1.04	0.90	0.60
25 mm	0.02	0.25	0.86	1.14	0.88	0.99	0.80
30 mm	0.07	0.32	0.99	1.16	0.93	1.08	0.85
50 mm	0.23	0.73	1.29	0.99	1.09	1.11	1.05

- The material shall be fibre free
- The density of the same shall be within 140-180 Kg/m³
- It shall have antimicrobial product protection, and shall pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM G 22, from an independent testing agency.
- The material shall have a thermal conductivity not exceeding 0.047 W/m.K @ 20 Deg. C
- The material shall withstand maximum surface temperature of +85°C and minimum surface temperature of -20°C
- The material shall conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & UL 94 (HBF, HF 1 & HF 2) in accordance to UL 94, 1996.
- The insulation shall pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12.7), from an independent testing agency.

Comply

Thickness of the material shall be as specified for the individual application. The insulation shall be installed as per manufacturer's recommendation.

DUCT ACOUSTIC LINING

Ducts so identified and marked on Drawings and included in Schedule of Quantities shall be provided with acoustic lining of thermal insulation material for a distance of minimum 5 meters (or 30% of the duct length whichever is more).

Installation Procedure

The inside surface for the ducts shall be covered with adhesive recommended by the manufacturer. Cut Foamed sheets into required sizes apply adhesive on the foam and stick it to the duct surface

Installation Procedure

The wall surface shall be cleaned and required surface preparation shall be done for applying adhesive. Rubber based contact adhesive recommended by the manufacturer shall be used. The foam sheets shall be cut to required size and a thin layer of adhesive shall be applied to both the surfaces; wall and i. When it is tack dry, it is applied / stuck with enough pressure to the walls/ceiling. Minimum 5 fasteners with washer (of G.I Sheet 2.5 inch x 2.5 inch) / square meter,



4 at corners & 1 at centre shall be put immediately after sticking with the help of adhesive. The length of the fastener shall be minimum 75 mm.

1. The fire-stop penetration seal for pipes:

The flexible fire stop pipe seal with a fire resistance of > 90 min. for non-combustible/combustible supply pipes and/or non-combustible waste pipes in solid ceilings/walls or light weight walls.

Create fire-resistant seals for pipework using closed-cell, flexible elastomeric insulation with intumescent effect

The material shall have following properties

Temperature Range: -50 Deg. C to +85 Deg.C

Thermal Conductivity: 0.05 W/m.K at 0 Deg.C

Water Vapour Diffusion Factor $\mu > 10,000$

Reaction to Fire: B2 in accordance to DIN 4102

Fire reaction to structural element: Minimum R 90

(in accordance to prEN 1366-3.2:N185 2007-07 and DIN EN 13501-2: 2008-01, section 7.5.8)

Installation:

The intumescent insulation is to be either sleeved over the pipe or slit and sealed using manufacturer's recommended adhesive. Longitudinal seams and butt joints are to be covered with nitrile rubber self-adhesive tape. The gap remaining between the insulation and the wall/ceiling is to be closed completely using mineral mortar. In lightweight walls, the gap is closed using loose fill within the wall and filler against the gypsum board (annular gap up to 50 mm).

The length of the intumescent insulation is to be determined depending on the pipe material and diameter according to manufacturer's recommendation as per the standard / test report for the R 90 test.

The Pipe diameters above 89 mm can be insulated with sheets which are additionally fixed with the binding wire (wound around 6 times per running meter). Binding of wire shall be done for tubes as well.

Installation with zero clearance between the insulated pipes is permitted in the area of penetration is permitted. In case of chilled water lines condensation aspect shall be considered.

Work shall be carried out in accordance with relevant R 90 testing standard / test report.

MODE OF MEASUREMENT:

- 1.0 Representatives from the Contractor, Client and Engineer shall conduct a joint inspection of the Equipments. All the discrepancies observed either incomplete works or defective work shall be clearly indicated in the joint inspection report. The mode of measurements given below is for the purpose of measurement and payment and the scope of works shall be as specified elsewhere in the specification. No additional payment shall be made for use of higher thicknesses of insulation than those specified, for whatever reason.
- 1.1 Thermal insulation for Duct/pipe
- Area of duct/pipe insulation finished as per specification shall be calculated on the basis of finished duct/pipe area before insulation based on the approved drawing / as built drawing.
- 1.2 Acoustic insulation for Duct
- Area of acoustic insulation finished as per specification shall be calculated on the basis of finished duct area before insulation based on the approved drawing / as built drawing and paid per unit area.

END OF SECTION 07

SECTION 08 CONDENSATE DRAIN PIPING AND REFRIGERANT PIPING DISTRIBUTION

1. SCOPE

This section lays down the general requirements for Supply, Installation and testing of all Piping works like Condensate drain piping and Refrigerant piping and related valves and accessories.

2. CODES AND STANDARDS

The material construction, manufacture, inspection, testing, and commissioning of water distribution piping shall comply with all currently applicable statutes, regulations and safety codes in the locality where the Equipment will be installed. Nothing in these specifications shall construe to relieve the CONTRACTOR of his responsibility. The equipment supplied shall comply with the latest applicable Indian and / or American Standards.

3. SCOPE OF SUPPLY AND ERECTION

3.1. The CONTRACTOR shall supply all piping material like pipes, fittings, hangers, miscellaneous steel required to support piping, flanges and other items as required.

3.2. Scope of erection to be performed by the CONTRACTOR is outlined below:

- a) The CONTRACTOR shall unload from carriers at plant site, handle, check, receive, transport, store, erect and test all materials furnished by him and others in accordance with this specification and General Conditions of Contract. The CLIENT shall be informed of any loss or damage within seven days of receipt of material.
- b) The CONTRACTOR shall also install small accessory piping and any specialties furnished for equipment such as relief valves, air vents, built-in bypass and other equipments of this type.
- c) CONTRACTOR's work includes the final tube or pipe connection at pumps, including the final connection with required supporting structure.
- d) The CONTRACTOR shall install primary elements for flow measurements, control valves and on-line metering equipment.
- e) The CONTRACTOR shall hydrostatically test the entire piping system including valves and specialties.
- f) All piping shall be internally cleaned, and flushed by the CONTRACTOR before and after erection in a manner suited to the service as directed by the CLIENT.
- g) For hydrostatic testing and flushing, the CONTRACTOR shall furnish necessary equipment and instruments, piping etc.
- h) The CONTRACTOR's scope under this includes the following:
 - i) All welding material and consumables required for the piping shall be included by the contractor
 - ii) Jointing material as required for all screwed joints. Fasteners (bolts, nuts, studs washers etc.) and gaskets is required for all flanged joints.